

Enterprise Data Architecture & Reconciliation

Financial Audit Dashboard Project Report

Project Domain	FinTech / Financial Analytics
Tools	Python, Pandas, SQLite, SQL, Regex, Matplotlib, Power BI
Reporting Period	Calendar Year 2023
Final Dataset	53,899 reconciled transactions
Total USD Revenue	\$148,274,622.91

This report documents the data extraction, cleaning, reconciliation, validation, and dashboard-ready output created for the CFO financial audit requirement.

1. Project Problem Statement

The project addresses a financial reporting and reconciliation problem in a FinTech environment. The CFO requires a Q1–Q4 financial audit dashboard, but the required information is distributed across three separate and inconsistent systems.

- **HR database:** historical user status records contain multiple records per user; only the latest status determines whether a user is Active.
- **Server logs:** 100,000 raw text records contain transaction fields inside strings; ERROR records must be removed and transaction fields extracted.
- **Finance exchange-rate data:** daily EUR-to-USD rates have missing weekend/market-closure values that must be handled without losing valid transactions.
- **Reporting:** the cleaned data must support monthly Total USD Revenue and Top 5 customer analysis.
- **Efficiency:** the pipeline must process the large log dataset using vectorized Pandas operations rather than row-by-row loops.

The solution therefore creates a reproducible Python pipeline that extracts, cleans, reconciles, validates, and prepares the financial data for interactive reporting.

2. Project Solution

Phase 1 — SQL Extraction

A SQLite window function ranks each user's historical records by updated_at. Only the latest record is retained and Active users are selected.

Phase 2 — Regex Data Extraction

Pandas string extraction with Regex parses Date, User ID, Product ID and Euro Value from the raw server logs after filtering ERROR records.

Phase 3 — Financial Reconciliation

Exchange-rate dates are sorted and missing market-closure rates are forward-filled. Transactions are merged by date and then matched to Active users by User ID.

Phase 4 — Revenue Calculation

USD Revenue is calculated as Euro Value multiplied by the applicable EUR-to-USD exchange rate.

Phase 5 — Validation & Dashboard Preparation

The final data is checked for missing values, duplicates, exchange-rate gaps, revenue totals and date range, then exported as a Power BI-ready CSV.

Tools Used

Python/Pandas for the pipeline; **SQLite/SQL** for user-status extraction; **Regex** for log parsing; **Matplotlib** for visual analysis; and **Power BI** for the interactive dashboard.

Source	Processing	Output
SQLite HR database	Latest status per user	Active users
Raw server logs	ERROR filtering + Regex	Clean transactions
EUR/USD CSV	Sort + forward-fill	Complete exchange rates
All cleaned sources	Date/User reconciliation	Final financial dataset
Final dataset	Aggregation + visualization	CFO dashboard-ready output

3. Project Report

The completed reconciliation produced a Power BI-ready transactional dataset covering the full 2023 reporting period. Each final record contains the transaction date, active customer, product, Euro value, applicable exchange rate, and calculated USD revenue.

Metric	Result
Final reconciled transactions	53,899
Unique customers represented	13,888
Total USD revenue	\$148,274,622.91
Average transaction value	\$2,750.97
Reporting period	2023-01-01 to 2023-12-31

The two primary dashboard outputs are monthly Total USD Revenue and the Top 5 Most Valuable Customers.

4. Project Screenshots

These visuals are generated directly from the completed project dataset and represent the project's report/dashboard outputs.

Screenshot 1 — Power BI-Ready Dataset

Power BI-Ready Dataset — Sample Records

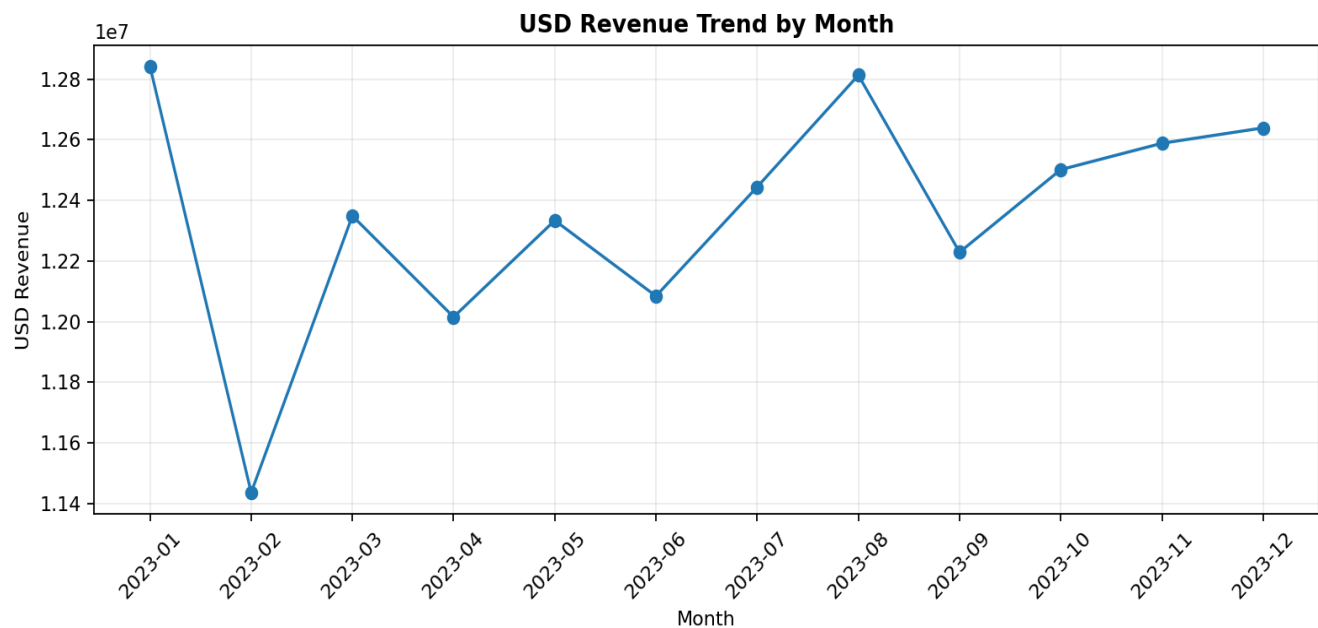
Date	User_ID	name	Product_ID	Euro_Value	EUR_to_USD	USD_Revenue
2023-03-15	U-19484	User_U-19484	PRD-874	3,404.44	1.1782	\$4,011.22
2023-10-21	U-15882	User_U-15882	PRD-284	164.08	1.0753	\$176.44
2023-01-10	U-22781	User_U-22781	PRD-665	26.80	1.1271	\$30.21
2023-10-20	U-22062	User_U-22062	PRD-363	48.63	1.0753	\$52.29
2023-12-02	U-22146	User_U-22146	PRD-223	960.60	1.0661	\$1,024.14
2023-03-21	U-08211	User_U-08211	PRD-741	1,129.34	1.0006	\$1,130.04
2023-02-01	U-12689	User_U-12689	PRD-223	3,390.14	1.1926	\$4,043.13
2023-07-18	U-06991	User_U-06991	PRD-732	690.08	1.1029	\$761.10

Screenshot 2 — Data Quality & Reconciliation Results

Project Data Quality & Reconciliation Results

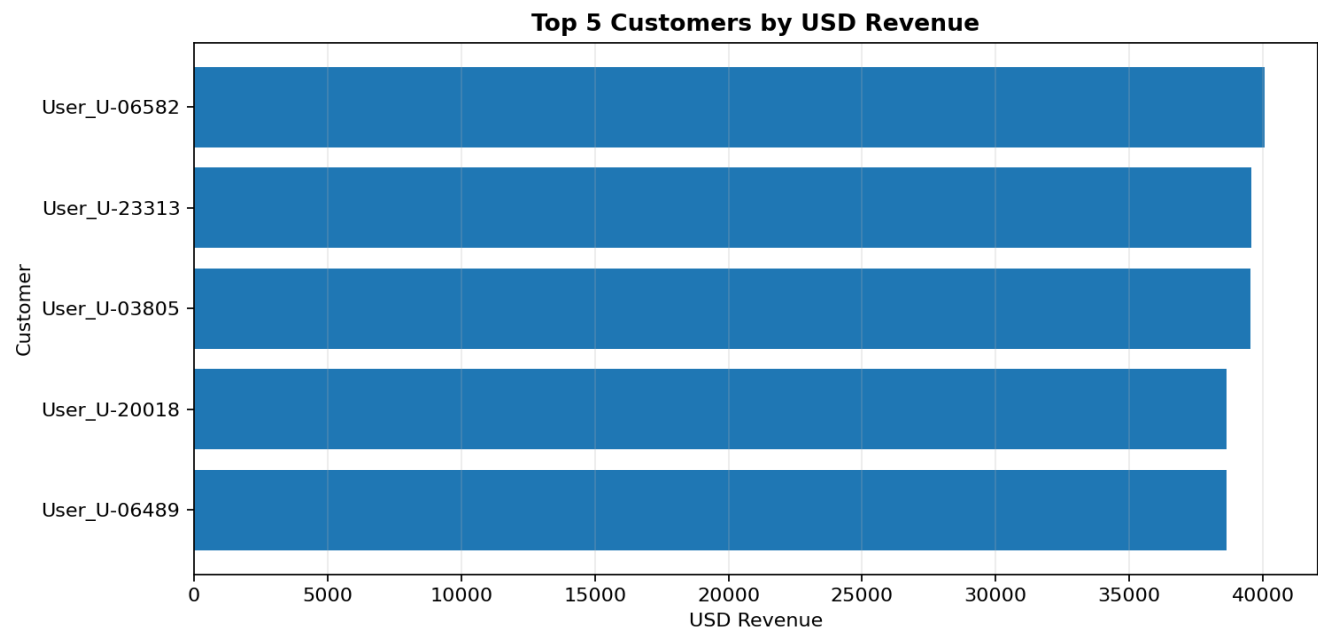
Validation Metric	Result
Final reconciled transactions	53,899
Unique customers	13,888
Total USD revenue	\$148,274,622.91
Average transaction	\$2,750.97
Date range	2023-01-01 to 2023-12-31
Missing USD revenue	0
Missing exchange rates	0

Screenshot 3 — Monthly USD Revenue Trend



This chart shows the month-by-month USD revenue trend across 2023.

Screenshot 4 — Top 5 Customers by Revenue



This chart identifies the five customers contributing the highest total USD revenue.

5. Conclusion

The project converts three separate and messy data sources into a single reconciled financial dataset suitable for audit reporting and business intelligence. SQL resolves the historical user-status problem, Regex converts unstructured server logs into structured transactions, forward-filling preserves transactions across missing exchange-rate dates, and Pandas performs the final reconciliation and aggregation.

The final dataset contains **53,899 reconciled transactions** and represents **\$148,274,622.91** in calculated USD revenue from 2023-01-01 to 2023-12-31. The resulting CSV can be loaded into Power BI for interactive date/quarter filtering, revenue KPIs, monthly revenue trends and Top 5 customer analysis.